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# Module 2.2 Aggregate Demand, Aggregate Supply and Equilibrium

## 1. Aggregate Demand (AD)

**Aggregate Demand (AD)** is the total demand for all goods and services in an economy at a given price level in a given time period. It represents the total planned expenditure in the economy.

$$AD = C + I + G + (X - M)$$

| Component | Definition | Typical % of UK GDP |
|---|---|---|
| **C — Consumption** | Household spending on goods and services | ~60-65% |
| **I — Investment** | Spending by firms on capital goods (machinery, equipment, buildings). Includes changes in inventories. | ~16-18% |
| **G — Government Spending** | Government expenditure on goods and services (not transfer payments like benefits) | ~20% |
| **(X – M) — Net Exports** | Exports minus imports. Positive net exports add to AD; negative net exports reduce AD. | Usually negative for UK (~-2 to -4%) |

### 1.1 Why the AD Curve Slopes Downward

The AD curve shows an inverse relationship between the price level and real GDP demanded. Three effects explain the downward slope:

- **Real balance (wealth) effect:** A higher price level reduces the real value of money and financial assets. Consumers feel poorer and spend less. C falls, so AD falls.
- **Interest rate effect:** A higher price level increases demand for money (to fund higher-priced transactions). This pushes up interest rates, which discourages borrowing and investment. I falls, so AD falls.
- **International trade effect:** A higher domestic price level makes exports more expensive and imports relatively cheaper. X falls and M rises, so net exports fall and AD falls.

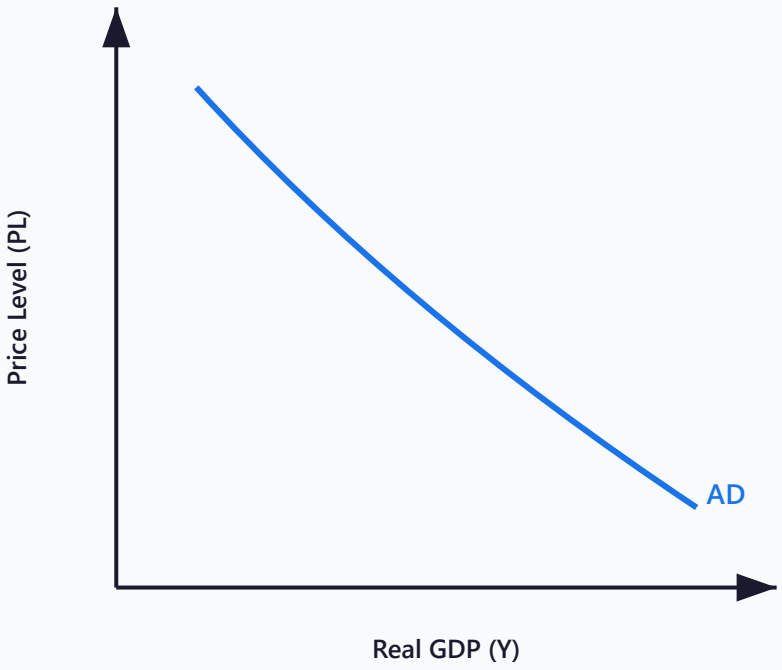


*Figure 1: The Aggregate Demand curve — downward sloping due to wealth, interest rate, and trade effects*

## 1.2 Factors that Shift the AD Curve

Any change in C, I, G, or (X-M) that is not caused by a change in the price level will shift the entire AD curve.

### Consumption (C) Shifters

- **Consumer confidence:** Optimism increases spending; pessimism reduces it.

- **Interest rates:** Lower rates reduce saving incentive and cost of borrowing, increasing C.
- **Wealth effects:** Rising house/share prices make consumers feel richer, increasing C.
- **Tax changes:** Income tax cuts increase disposable income, increasing C.
- **Availability of credit:** Easier credit access increases C.
- **Income distribution:** Redistribution to lower-income groups (higher marginal propensity to consume) increases C.

### Investment (I) Shifters

- **Interest rates:** Lower rates reduce the cost of borrowing for investment.
- **Business confidence/expectations:** Optimism about future demand increases I.
- **Corporation tax:** Lower rates increase post-tax returns on investment.
- **Technological change:** New technologies create investment opportunities.
- **Retained profits:** Higher profits provide internal funds for investment.

### Government Spending (G) Shifters

- **Fiscal policy decisions:** Government can increase or decrease spending on public services, infrastructure, defence.
- **Political priorities:** Elections, public pressure, economic conditions influence spending decisions.

### Net Exports (X – M) Shifters

- **Exchange rate:** Depreciation makes exports cheaper and imports dearer, improving net exports.
- **Overseas income levels:** Growth in trading partners increases demand for UK exports.
- **Relative inflation:** Lower UK inflation improves competitiveness.
- **Trade policy:** Tariffs, quotas, trade agreements affect flows.

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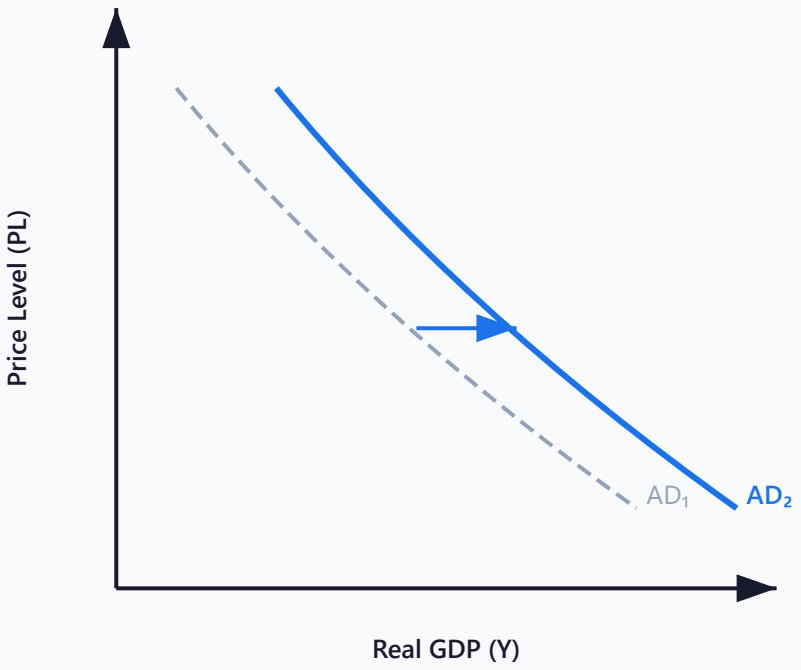


*Figure 2: Rightward shift in AD — caused by an increase in C, I, G, or net exports*

# 2. Aggregate Supply (AS)

## 2.1 Short-Run Aggregate Supply (SRAS)

**Short-Run Aggregate Supply (SRAS)** shows the total output that firms in the economy are willing and able to supply at each price level in the short run, when some factor prices (especially wages) are sticky/fixed.

The SRAS curve slopes **upward** because in the short run:

- Wages and some input costs are sticky (fixed by contracts) — when the price level rises, firms' revenues increase but costs lag behind, so profit margins rise, incentivising greater output.
- Higher prices signal higher demand, encouraging firms to produce more.

### Factors that Shift SRAS

Any change in the costs of production shifts the SRAS curve:

- **Wage costs:** Wage increases shift SRAS left (higher costs).
- **Raw material/energy prices:** Oil price rises shift SRAS left.
- **Exchange rate:** Depreciation raises import costs, shifting SRAS left.
- **Indirect taxes:** Increases shift SRAS left; reductions shift right.
- **Subsidies:** Increases shift SRAS right.
- **Productivity:** Improved productivity reduces unit costs, shifting SRAS right.

## 2.2 Long-Run Aggregate Supply (LRAS)

### The Classical (Monetarist) View

**Classical LRAS** is a **vertical line** at the full-employment level of output (Yfe). In the long run, the economy always returns to full employment. Output is determined solely by supply-side factors (quantity and quality of factors of production), not by the price level or demand.

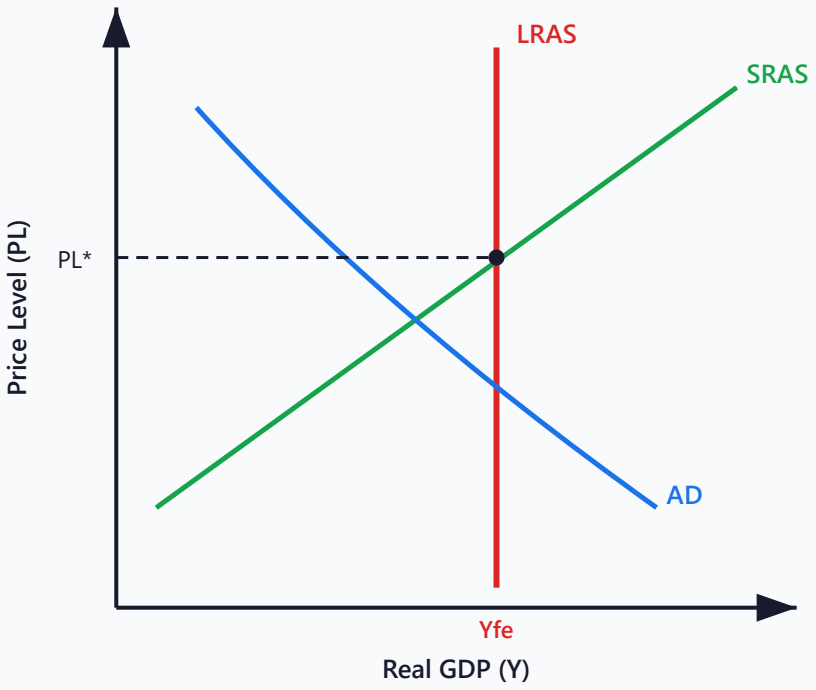


*Figure 3: Classical/Monetarist model — vertical LRAS at full-employment output (Yfe)*

In the Classical view:

- An increase in AD only raises the price level in the long run — it does not increase real output.
- The economy self-corrects: if AD falls causing unemployment, wages fall, costs decrease, SRAS shifts right, and the economy returns to Yfe at a lower price level.
- Long-run growth requires shifting LRAS rightward through supply-side improvements.

## Factors that Shift Classical LRAS

- Increase in quantity/quality of **labour** (immigration, education, training)
- Increase in **capital stock** (investment in machinery, infrastructure)
- **Technological progress** (innovation, automation)
- Discovery of **natural resources**
- **Institutional improvements** (property rights, rule of law, deregulation)

## The Keynesian View

**Keynesian LRAS** has **three distinct sections**: a horizontal section (spare capacity — output can increase without inflation), an upward-sloping section (approaching full capacity — increasing costs and bottlenecks), and a vertical section (full capacity — no more output possible).

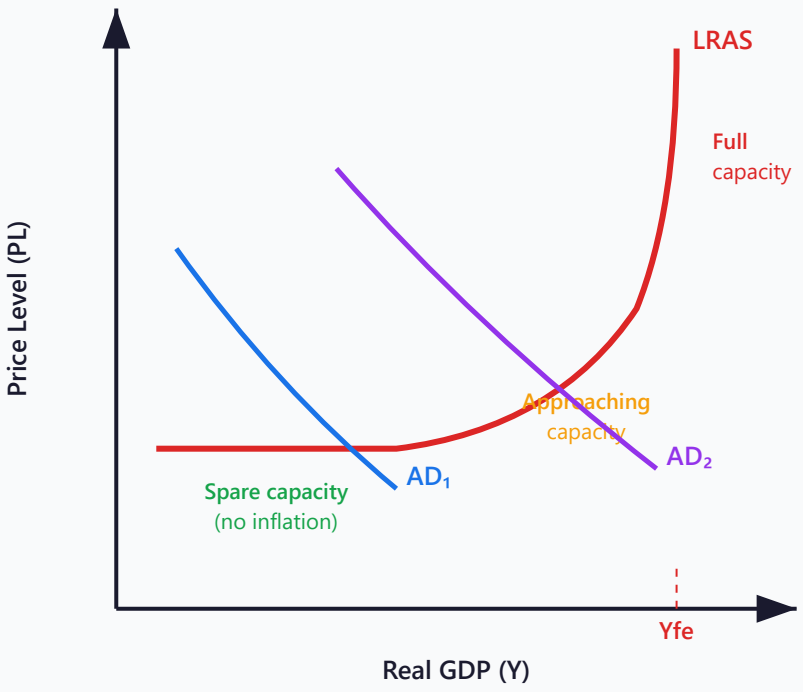

*Figure 4: Keynesian LRAS — three sections reflecting spare capacity, approaching capacity, and full capacity*

Key Keynesian insights:

- The economy can settle in a **deflationary gap** (below full employment) for a prolonged period — it does not self-correct automatically.
- In the spare-capacity section, increasing AD boosts output without causing inflation (justifying demand management).
- Only when the economy approaches full capacity does inflation become a problem.
- Government intervention (fiscal policy) is needed to close deflationary gaps — the market will not self-correct in the short to medium run.

**Case Study: Keynesian vs Classical — Response to 2008 Financial Crisis**

The 2008 crisis tested both schools of thought. Keynesian economists argued for fiscal stimulus — government spending to replace collapsed private demand. The UK initially followed this approach (bank bailouts, temporary VAT cut from 17.5% to 15%, £20bn fiscal stimulus). Classical/monetarist economists warned that government spending would crowd out private investment and lead to unsustainable debt. From 2010, the Coalition government shifted to austerity (spending cuts, tax rises to reduce the deficit) — more aligned with the Classical view. Keynesians argued this prolonged the recession. The debate continues: UK GDP took five years to recover to pre-crisis levels, and critics argue austerity slowed the recovery.

# 3. AD/AS Equilibrium and Analysis

**Macroeconomic equilibrium** occurs where AD intersects AS. In the short run, equilibrium is where AD = SRAS. In the long run (Classical model), the economy settles where AD = SRAS = LRAS at the full-employment output level.

## 3.1 Demand-Pull Inflation (AD/AS Analysis)

An increase in AD (rightward shift) along an upward-sloping SRAS or in the intermediate range of Keynesian LRAS causes both real GDP and the price level to rise — **demand-pull inflation**.

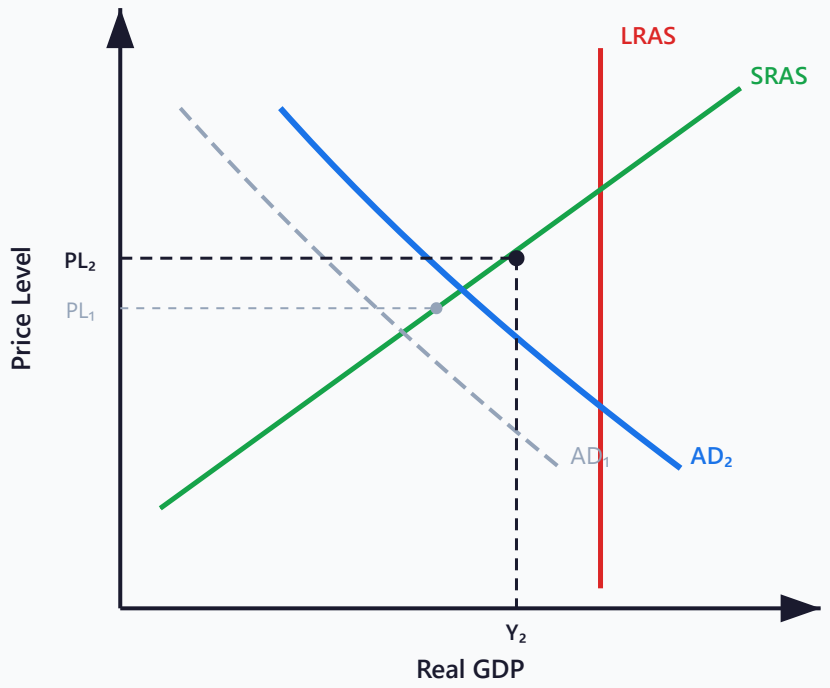


*Figure 5: Demand-pull inflation — AD shifts right, raising price level and real GDP in the short run*

## 3.2 Cost-Push Inflation (AD/AS Analysis)

A decrease in SRAS (leftward shift) causes the price level to rise while real GDP falls — **cost-push inflation** (stagflation).

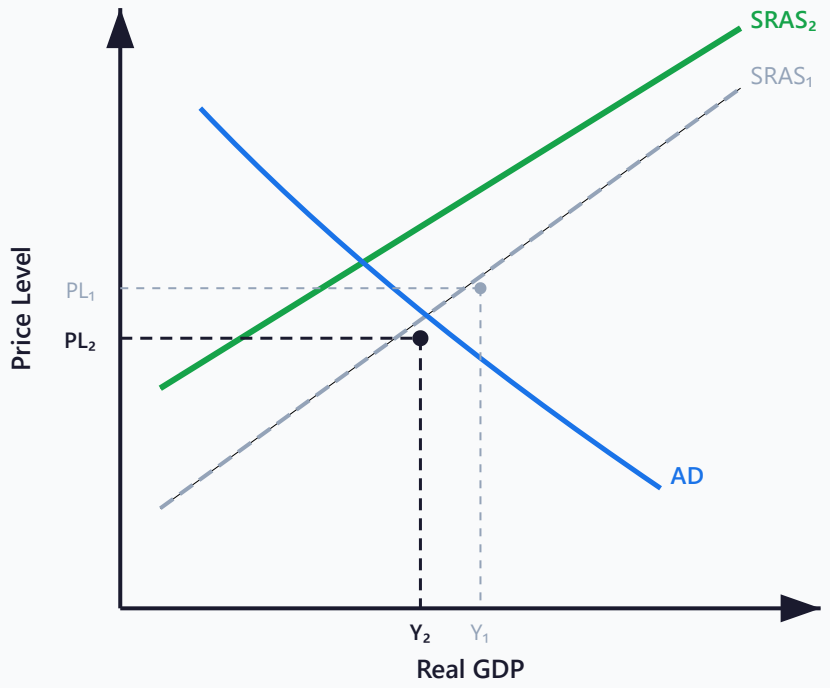


*Figure 6: Cost-push inflation — SRAS shifts left, price level rises while real GDP falls (stagflation)*

### Case Study: 1970s Oil Price Shocks — Stagflation

The 1973 OPEC oil embargo quadrupled oil prices, and the 1979 Iranian Revolution caused a further doubling. As oil is a key input for virtually all production, this caused a massive leftward shift of SRAS across Western economies. The result was stagflation — simultaneous inflation and recession. UK inflation reached 24% in 1975 while unemployment rose sharply. This posed a major policy dilemma: expansionary policy to combat unemployment would worsen inflation, while contractionary policy to fight inflation would deepen the recession. The experience led to a rethinking of Keynesian demand management and contributed to the rise of monetarism and supply-side economics.

## 3.3 Long-Run Economic Growth (AD/AS Analysis)

Long-run growth is shown by a rightward shift of the LRAS curve (Classical) or an extension of the Keynesian LRAS. This increases the economy's productive capacity, allowing higher output without inflationary pressure.

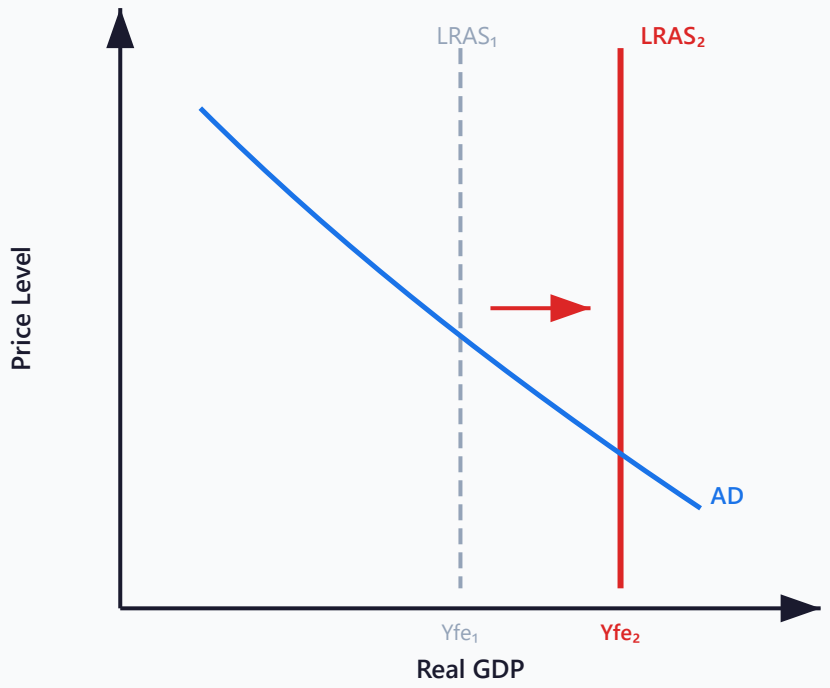


*Figure 7: Long-run economic growth — LRAS shifts right, increasing potential output*

## 4. The Multiplier Effect

**The multiplier effect** describes how an initial injection into the circular flow of income leads to a larger final increase in national income (GDP). An initial increase in spending creates income for others, who then spend a proportion of it, creating further income, and so on — a chain reaction.

$$\text{Multiplier (k)} = 1 / (1 - \text{MPC}) = 1 / \text{MPW}$$

Where:

- **MPC** = Marginal Propensity to Consume (fraction of extra income that is spent domestically)
- **MPW** = Marginal Propensity to Withdraw = MPS + MPT + MPM (saving + tax + imports)

$$\text{Change in GDP} = \text{Multiplier} \times \text{Initial injection}$$

## 4.1 How the Multiplier Works

Example: Government spends an extra £1 billion on infrastructure. If MPC = 0.8 (people spend 80% of extra income):

- Round 1: £1bn spent → becomes income for construction workers
- Round 2: Workers spend 80% = £0.8bn → becomes income for shops, suppliers
- Round 3: Those recipients spend 80% = £0.64bn
- And so on... Total increase in GDP = £1bn × (1/0.2) = £5bn

The multiplier is **larger** when MPC is high (low savings rate, low tax rate, low import propensity).

The multiplier is **smaller** when withdrawals are high (high savings, high taxes, high import spending).

## 4.2 The Negative Multiplier

The multiplier works in reverse too. A withdrawal of spending (e.g. government austerity, fall in investment) causes a multiplied contraction in GDP. This is why recessions can deepen rapidly.

## 4.3 Limitations of the Multiplier

- Assumes spare capacity exists — if the economy is at full employment, extra spending just causes inflation, not real output growth.
- Time lags — the multiplier effect takes time to work through the economy (months or years).
- Crowding out — government borrowing to fund spending may raise interest rates, reducing private investment (offsetting the multiplier).
- Open economy leakages — in a globalised economy, spending leaks abroad through imports, reducing the domestic multiplier.
- Consumer and business confidence — if confidence is low, people may save extra income rather than spend it (MPC falls).
- Difficult to measure precisely — the actual MPC and multiplier are uncertain and vary over time.

### Case Study: COVID-19 Furlough Scheme and the Multiplier

The UK's Coronavirus Job Retention Scheme (furlough) paid up to 80% of wages for employees

who could not work during lockdowns. By supporting approximately 11.7 million jobs at a cost of £70 billion, the scheme prevented a massive negative multiplier effect. Without it, mass unemployment would have caused a collapse in consumer spending, which would have cascaded through the economy as businesses lost customers, leading to further redundancies and spending cuts. The IMF estimated that UK GDP would have fallen significantly more than the 9.7% recorded in 2020 without the scheme. However, critics argued the scheme also supported some "zombie" jobs in firms that would not survive long-term.

**Exam tip:** Always use AD/AS diagrams to illustrate macroeconomic analysis. When discussing policy, show the shift in AD or AS, identify the new equilibrium, and explain the impact on both the price level and real GDP. Consider both short-run and long-run effects, and evaluate using the multiplier where appropriate.

**Evaluation — Classical vs Keynesian**

- **Classical/monetarist view:** The economy self-corrects to full employment in the long run. Government intervention is unnecessary and may cause inflation. Focus on supply-side policies to shift LRAS right.
- **Keynesian view:** The economy can remain stuck below full employment for extended periods. Demand management (fiscal policy) is needed to close output gaps. The multiplier means fiscal policy can be powerful.
- **In practice:** Most governments use a combination — demand management in recessions (Keynesian) and supply-side reforms for long-run growth (Classical). The debate is about the balance and timing of intervention.
- **Key question for evaluation:** How long is the "long run"? If markets take years to self-correct, the human cost (unemployment, lost output) may justify Keynesian intervention even if the Classical view is ultimately correct.

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